

1. Applying for IITD-HPC

1.1. Go to <https://userm.iitd.ac.in/usermanage/hpc.html>

1.2. The Webpage below will open, fill in the details.

HPC facility, CSC, IIT Delhi

Login page for HPC usage request and Project Management

Enter Your Login Information:

Username

Password

Login To The System

1.3. Once login, it will ask for supervisor uid and expiry date, fill as required. An email will be received once approved by the supervisor.

Welcome *Rajan*

Application for HPC access

If you want to use HPC, please submit the following information to complete your request:

Faculty supervisor (uid):

Requested Expiry date of access: 

Request Access

Logout

1.4. The supervisor has to add a project with some budget.

2. Setup HPC account (to be done only once):

2.1. Login

```
ssh kerberosID@hpc.iitd.ac.in
```

enter your kerberos password

2.2. Setup environment (followed using <https://supercomputing.iitd.ac.in/?access>)

```
cp /home/apps/skeleton/.bashrc $HOME/.bashrc
```

```
cp /home/apps/skeleton/.bash_profile $HOME/.bash_profile
```

```
ln -s $SCRATCH $HOME/scratch
```

```
chmod og-rx $SCRATCH
```

```
chmod og-rx $HOME
```

2.3. Generate SSH key (followed using <https://supercomputing.iitd.ac.in/?sshkeys>)

```
ssh-keygen -t rsa
```

Note: Keep clicking Enter for all the options that appears (default options)

3. Using HPC

3.1. Login

```
ssh kerberosID@hpc.iitd.ac.in
```

enter your kerberos password

3.2. Running a job

IITD currently uses qsub as a job scheduler. Once logged in, and all scripts and data are at home, a job can be executed in two modes.

3.2.1. Interactively (while developing a pipeline).

```
qsub -I -P project_name -l select=1:ncpus=8:ngpus=0:mem=8G -l  
walltime=00:00:1
```

3.2.2. Directly submit (when pipeline is ready and can directly be executed).

This required a script to be made that contained all the instructions. Once the job is finished, an email will be sent to inform about job completion.

```
qsub -P project_name myjobscript
```

3.3. Checking status of running job

```
qstat -u $USER
```

3.4. Killing a job

```
qdel job_id
```

3.5. Check storage quota

```
lfs quota -hu $USER /home
```

```
lfs quota -hu $USER /scratch
```

3.6. Transfer data

```
scp -r lab1_data/ kerberosID@hpc.iitd.ac.in:/path/to/dir/
```